$$= \int \sqrt{(\cos x + \sin x)^2}\, dx = \int (\cos x + \sin x) dx$$

$$= \sin x - \cos x + c$$

Example 11.25

Evaluate : $\int \frac{x^3+2}{x-1} dx.$

Solution

$$\int \frac{x^3+2}{x-1} dx = \int \frac{x^3-1+3}{x-1} dx = \int \left(\frac{x^3-1}{x-1} + \frac{3}{x-1} \right) dx$$

$$= \int \left[\frac{(x-1)(x^2+x+1)}{x-1} + \frac{3}{x-1} \right] dx$$

$$= \int \left(x^2 + x + 1 + \frac{3}{x-1} \right) dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + x + 3\log|(x-1)| + c.$$

Example 11.26

Evaluate : (i) $\int a^x e^x dx$ (ii) $\int e^{x\log 2} e^x\, dx.$

Solution

(i) $$\int a^x e^x dx = \int (ae)^x dx = \frac{(ae)^x}{\log(ae)} + c$$

(ii) $$\int e^{x\log 2} e^x dx = \int e^{\log 2^x} e^x dx = \int 2^x e^x dx$$

$$= \int (2e)^x dx = \frac{(2e)^x}{\log(2e)} + c.$$

Example 11.27

Evaluate : $\int (x-3)\sqrt{x+2}\, dx.$

Solution

$$\int (x-3)\sqrt{x+2}\, dx = \int (x+2-5)\sqrt{x+2}\, dx$$

$$= \int (x+2)\sqrt{x+2}\, dx - 5\int \sqrt{x+2}\, dx$$

$$= \int (x+2)^{\frac{3}{2}} dx - 5\int (x+2)^{\frac{1}{2}} dx$$

$$= \frac{(x+2)^{\frac{5}{2}}}{\frac{5}{2}} - 5\frac{(x+2)^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$= \frac{2}{5}(x+2)^{\frac{5}{2}} - \frac{10}{3}(x+2)^{\frac{3}{2}} + c.$$

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